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Summary

Performance-driven **Backend Developer** with expertise in designing **scalable, real-time, and event-driven distributed systems** using Node.js, Express, and MySQL. Specialized in managing **high-concurrency**, **transactional integrity (ACID)**, and asynchronous processing using message queues and caching layers.

Technical Skills

Backend: Node.js, Express.js, RESTful API Design
Database: MySQL , MongoDB, Redis
Security: JWT Authentication, HTTP-only Cookies, RBAC, Rate Limiting
Tools & Languages: JavaScript (ES6+), Java, C, C++, Git, Postman

Experience

Full Stack Developer Intern — Jun 2025 – Nov 2025
Infinova Global — Remote

- Engineered responsive frontend interfaces using **React.js** and **Tailwind CSS**
- Integrated REST APIs with secure authentication and centralized error handling protocols
- Implemented secure **JWT-based authentication** workflows utilizing HTTP-only cookies
- Developed real-time data synchronization features using **WebSockets**
- Collaborated on backend architecture to ensure high system consistency and reliability

Projects

CardioGuard — Real-Time Health Monitoring System
Node.js, Express.js, MySQL, Redis, Socket.io, BullMQ

- **Architected** a real-time health monitoring system supporting **1000+ concurrent users** with continuous vitals streaming
- Designed an **event-driven, asynchronous processing pipeline** using Redis and BullMQ, enabling non-blocking request handling and improved system throughput
- Implemented **distributed worker architecture** with concurrency control and horizontal scaling, increasing processing capacity from **150 req/sec to 250+ req/sec**
- Conducted **load testing using k6**, simulating up to **1000 concurrent users**, identifying performance bottlenecks and optimizing system stability under high load
- Identified and resolved system bottlenecks by optimizing worker concurrency and introducing queue-based processing for high-throughput scenarios
- Built an intelligent alert detection engine using rolling vitals analysis with cooldown logic, reducing false alerts by **40%**
- Developed a scalable multi-recipient notification system delivering alerts to patients, families, and hospitals with fault-tolerant retry mechanisms
- Implemented geo-proximity hospital routing using the **Haversine formula**, reducing emergency assignment time to **<1 second**
- Ensured **data consistency under high concurrency** using MySQL transactions and row-level locking

SecureSend — Transactional Wallet System
Node.js, Express.js, MySQL

- **Engineered** a high-concurrency financial system supporting simultaneous transactions with **zero data inconsistency**
- Implemented a **double-entry ledger** ensuring **100% financial accuracy** and auditability across all transactions
- Designed **idempotent APIs** reducing duplicate transaction risk by **90%** in retry scenarios
- Handled deadlocks and failures using retry mechanisms, improving transaction success rate under load

- Secured financial endpoints with JWT authentication and rate limiting to prevent abuse and unauthorized access

Infinova CMS — Backend System
Node.js, Express.js, MongoDB

- Designed REST APIs handling dynamic content operations with efficient CRUD workflows
- Structured MongoDB schemas improving query performance and reducing response time by **30%**
- Built backend for admin dashboard supporting **multiple content types** (courses, testimonials, services)
- Ensured modular and scalable backend architecture enabling easy feature extensions

Education

Bachelor of Engineering in Computer Engineering 2022 – 2026
Sinhgad Institute of Technology **CGPA: 8.31**